Cereal 7 — Known Issues and Gotchas

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One bug in `train_specialist()` likely affects every saved specialist and its OOF predictions; fix it before re-running anything. The rest are evaluation weak spots and errors that were already solved.

Bugs and evaluation weak spots

Where	Problem	Effect	Fix
`train_specialist()`	Saves the model as it stands after phase 2B and then deletes the best-weights checkpoint. Early stopping only restores the best epoch within each phase.	When an earlier phase was better, the saved model and its OOF predictions are the worse one. S5 is a confirmed case (0.8026 frozen vs 0.7919 after 2B).	Track the best validation AUC across all three phases; save and predict with that model; keep the checkpoint file.
Specialist training data	S1–S4 train only on images labelled healthy or their one disease.	They never see the 38% of images with several labels, which the test folds contain.	Train one-vs-rest on all images, or include multi-disease images as positives.
Threshold tuning	Thresholds chosen on the same OOF predictions that are then scored.	Tuned F1 is optimistic.	Tune on an inner validation split.
No duplicate screen	Maize was never run through the MD5 + dHash screen.	Unknown leakage between folds.	Screen and use grouped multilabel folds.
Old summary PDF	Listed Other = 136 and UnidentifiedDisease = 201, and per-label F1 values marked approximate.	Wrong counts in circulation.	Use `Database.csv`: 324 and 332.
8-label OOF assembly (section 14)	Images outside a binary specialist's training subset get prediction 0 for that disease, but at inference (section 16) every image goes through every specialist.	The OOF score does not measure the deployed pipeline; a GLS specialist that never saw NCLB-only leaves may fire on them in use.	Score each specialist on every image in its held-out fold, matching inference.
SR label	39 positives in 2,355 images.	Any SR number is noisy (about 8 positives per test fold).	Report SR separately; don't let it drive design decisions alone.

Errors already solved

Error	Cause	Fix
`AttributeError: Normalization has no .layers`	ConvNeXt-Base inserts a Normalization layer at index 1, so the backbone isn't where code expects	`get_backbone()` scans `model.layers` for the first sub-model with `.layers`
`ValueError: array is read-only` in the co-occurrence cell	The NumPy array taken from the co-occurrence table is read-only	`co.to_numpy(dtype=float).copy()` before `np.fill_diagonal()`
`ModuleNotFoundError: iterstrat`	Wrong pip package name	`pip install iterative-stratification`
Rare labels over-predicted at 0.5	Weighted BCE favours recall	Per-label threshold tuning
Long runs dying mid-way	Session limits / crashes	`specialist_checkpoint.json` + per-fold OOF `.npy` + ModelCheckpoint